LangChain

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LangSmith Self-hosted

Deployment, Adoption, and Scale.

Tom Straub • Architect • Tuesday, October 27th, 2026

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Today

Agenda

- LangSmith Platform Architecture
- Terraform Module Review Agent Development
- Friction Log
- Q & A

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Platform Architecture

Self-hosted LangSmith on a dedicated GKE cluster.

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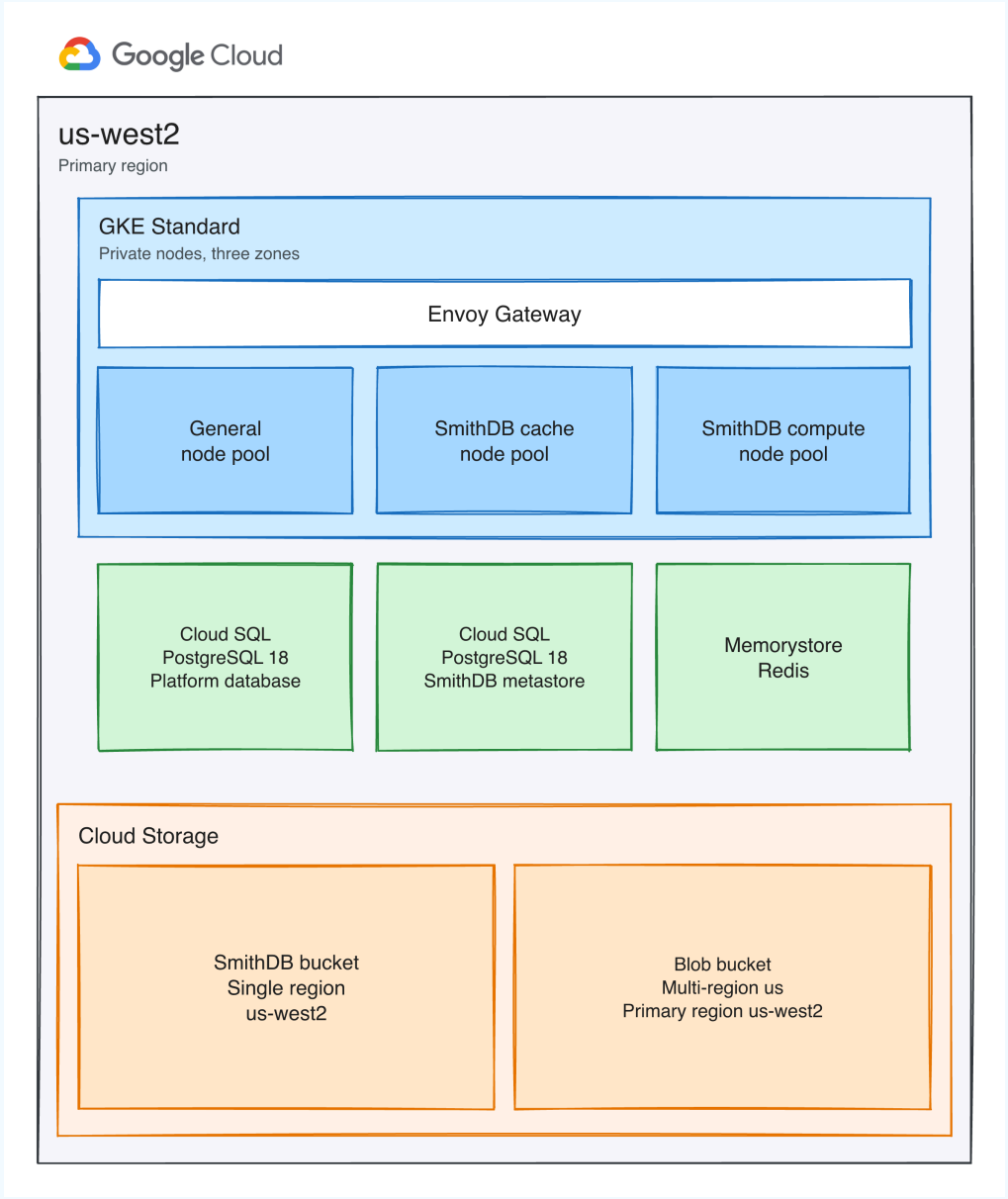
Platform

What we designed around

- A dedicated GKE Standard cluster.
- Agents stay where they are and send traces to LangSmith.
- Durable data sits in managed services, outside the cluster.
- A standard install now. Hosted agents come later.

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Architecture



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Platform

Infrastructure

Application

A dedicated GKE Standard cluster.

- General node pool
- SmithDB Cache node pool
- SmithDB Compute node pool

Storage

- Cloud SQL - Users, projects, datasets, experiments, and Chat.
- Blob GCS bucket - Payloads and attachments.
- Redis - Ingest queue, cache, and Chat.
- SmithDB Cloud SQL - Metastore, empty and separate.
- SmithDB GCS bucket - Traces and feedback.

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Platform

Scaling

The size we start at

About 20 concurrent users and 100 traces a second, with bursts toward 1,000

The size we move to

About 50 concurrent users and 1,000 traces a second, when the first ranges stay full.

- Redis - Grows with users.
- Blob GCS bucket - Grows with payloads.
- SmithDB Cloud SQL - A larger metastore.
- SmithDB Cache and compute - Larger machines.
- SmithDB GCS bucket - Grows with retained traces.

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BYOC

If operating LangSmith yourself is more than you want, we can talk about "Bring Your Own Cloud"

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Agent Development: Terraform Module Review

A module review that stops before it opens a pull request.

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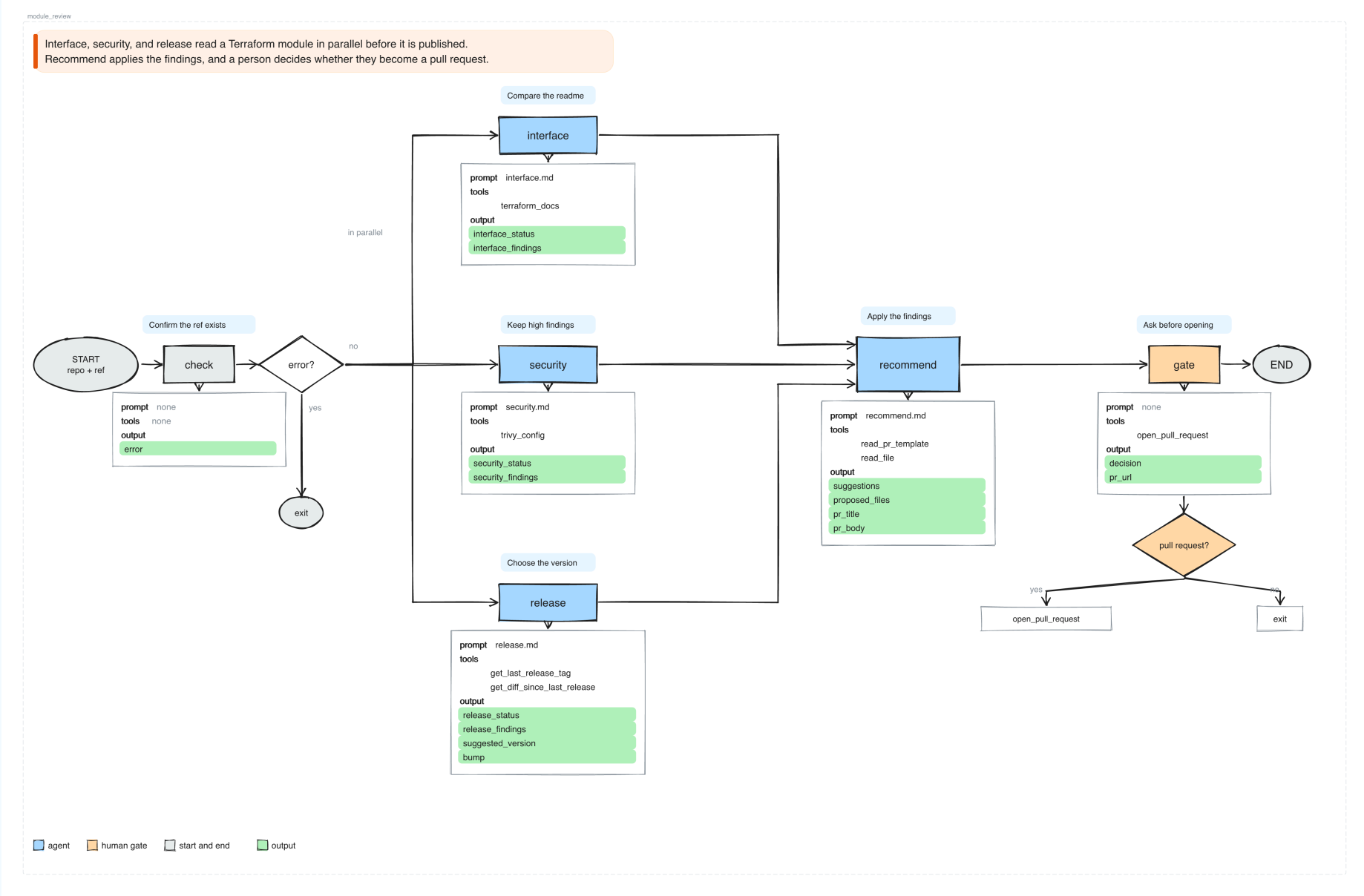
Agents

What Was Built

- The work is a Terraform module that is ready to publish.
- Interface, security, and release read the repo in parallel.
- Recommend drafts the files and the pull request.
- A person answers yes or no before a Pull Request is created.

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Agent Architecture



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Agents

Evaluation Results

Datasets

Seven frozen modules.

Repeatable scenarios.

Live data loop from real runs.

Evaluators

5 scores from the Code Evaluator.

LLM as a Judge.

Experiments

Prompt tuning and agent architecture changes.

Becomes a valuable development loop.

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Agents

Business Value

Self-service

The module docs are how teams learn to build infrastructure.

Clear module interfaces are what let that scale.

Mistakes in the module docs become tickets on the next release.

The right bump

Major, minor, and patch are applied consistently

Every candidate gets the same recommendation from the org rule, before the tag.

Flexibility and Scale

Recommendations can expand and improve over time.

Consistent recommendations across a large organization.

Human in the Loop gives teams a chance to engage with the recommendation.

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Friction

Lessons learned.

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Friction

Lessons Learned

- The eval stops at the human gate so the pull request is never scored.
- Agent development mistakes cost tokens and time.
- Starting with Evaluations and Experiments would have been better.

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Friction

What we would ask for

- Ability to set max tokens and interrupt.
- Automatically stop tool calls when you get the same result twice (same input).
- Template syntax for User Prompts.

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Questions?

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Thank you

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